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Dynamic Heterogeneous Graph Representation Learning: A Survey

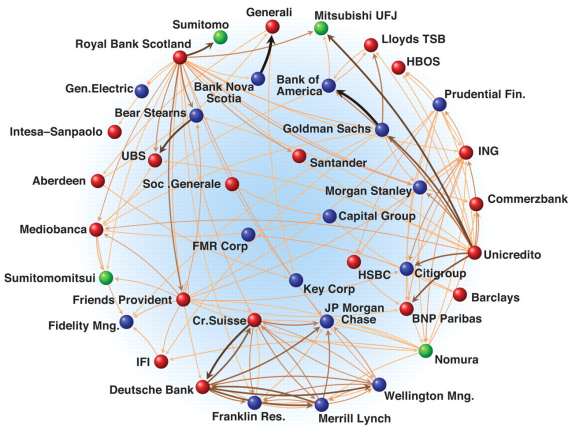
Huan Liu[1], Pengfei Jiao[1*], Jie Yin[2], Hongjiang Chen[1], Zhidong Zhao[1,3]

[1]Hangzhou Dianzi University, China [2]The University of Sydney, Australia
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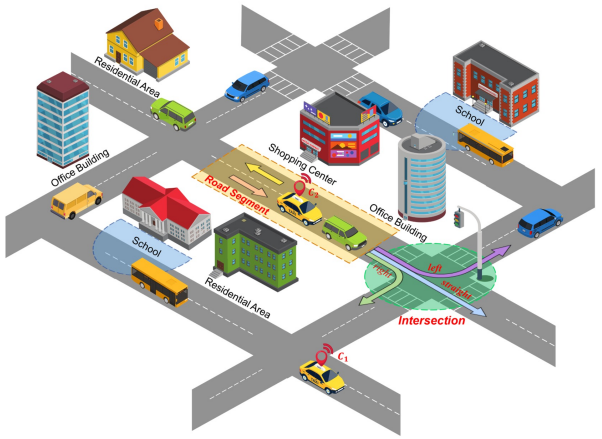
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Huan Liu, Pengfei Jiao*, et al. Dynamic Heterogeneous Graph Representation Learning: A Survey. IJCAI Survey Track, 2026.

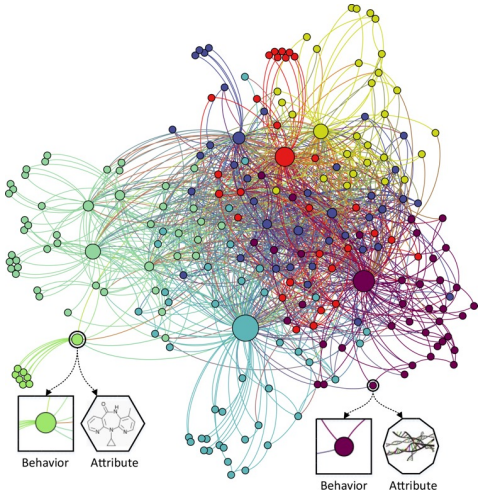
Background



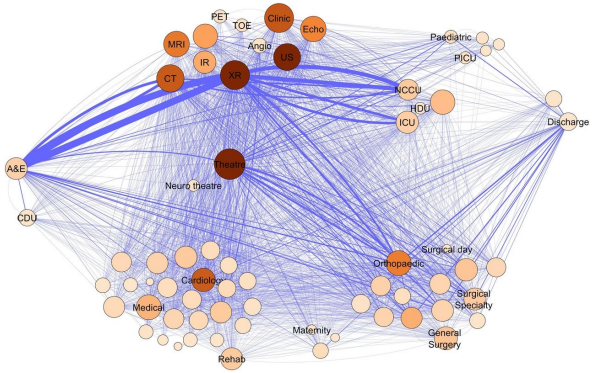
Economic network



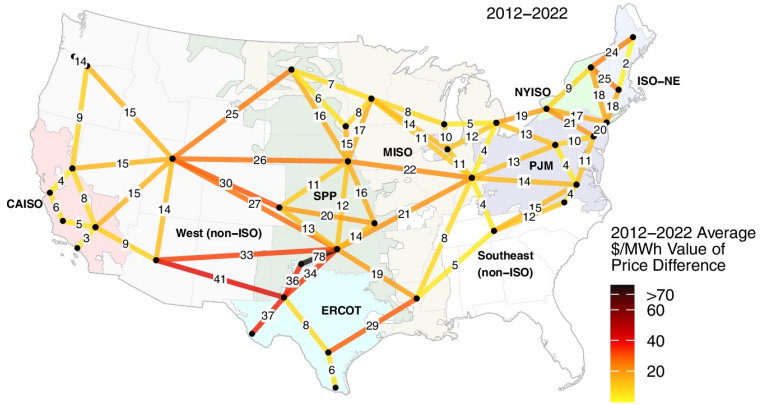
Traffic network



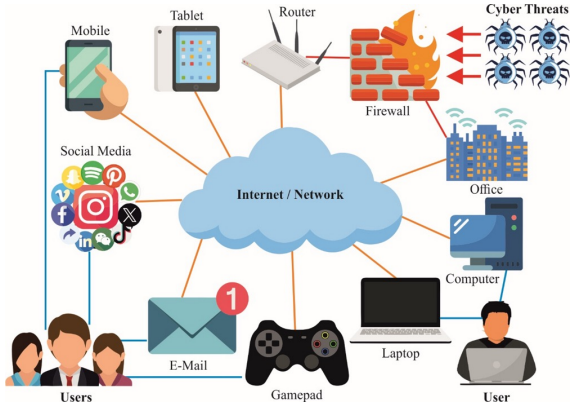
Biological network



Healthcare network

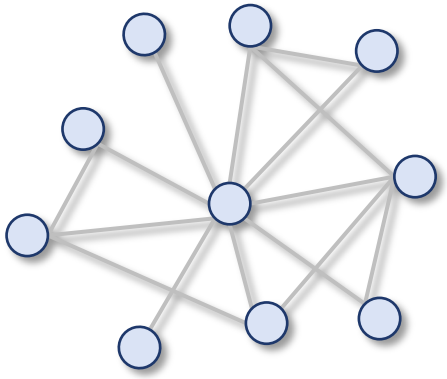


Energy network

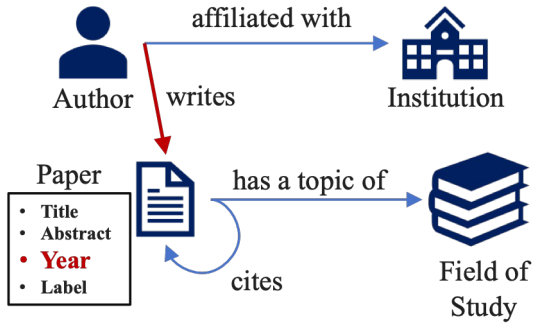


Internet network

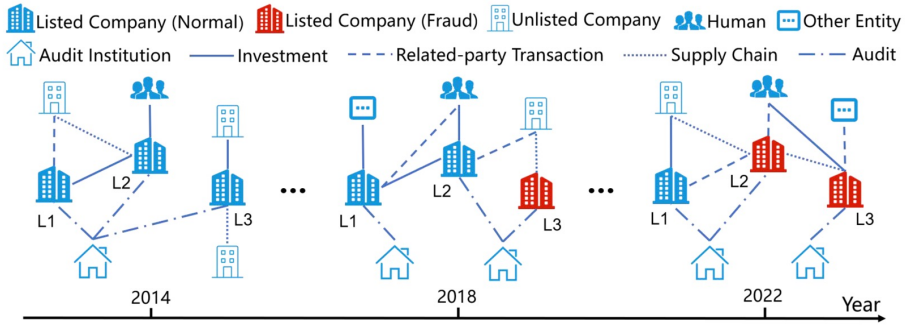
Background



Static, Homogenous Graph

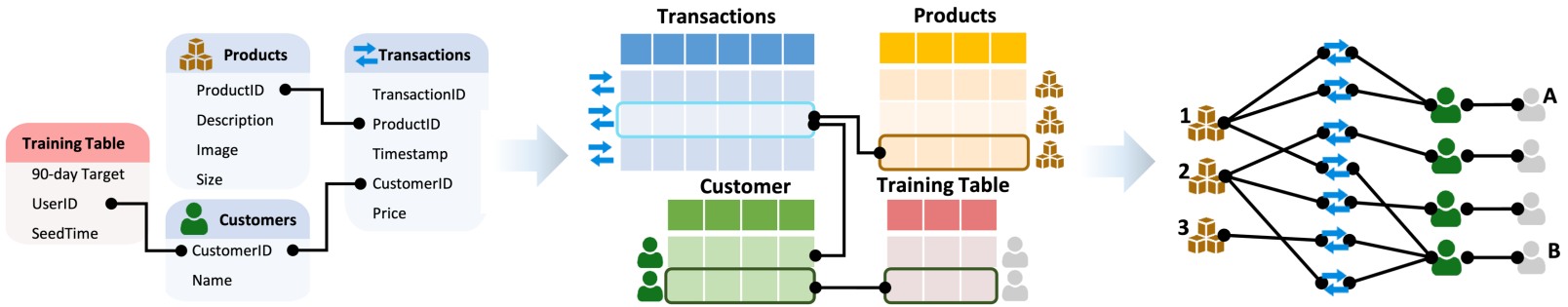


Academic [Wang et al, 2020]

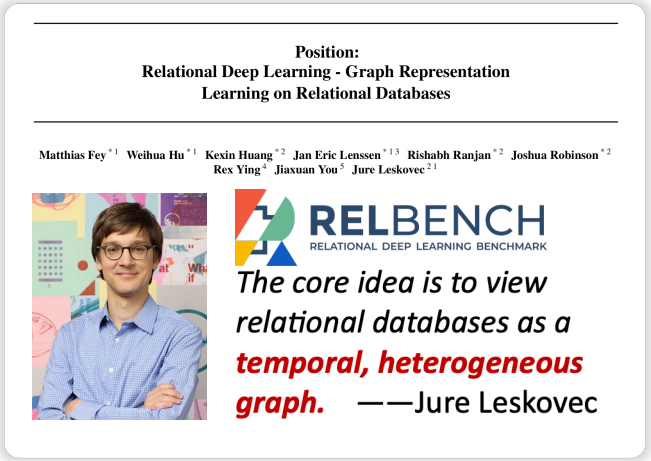


Financial [Wang et al, 2025]

Dynamic Heterogeneous Graph (DHG)



Deep Learning Pipeline for Relational Database [Dwivedi et al, 2025]

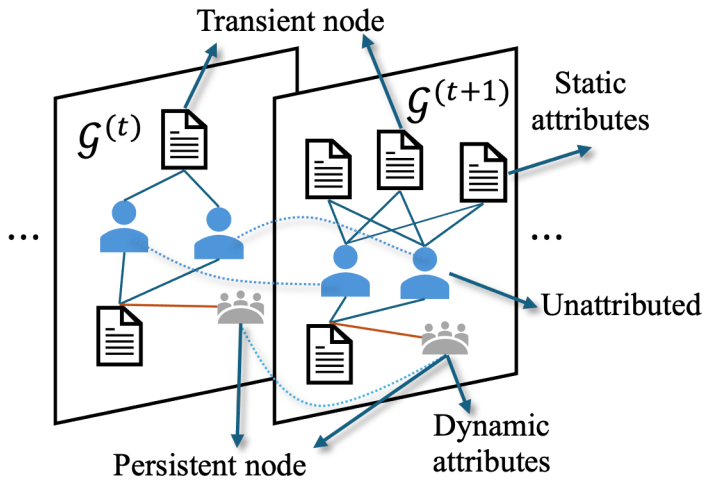


Most real-world systems are simultaneously **heterogeneous** and **dynamic**

Challenge

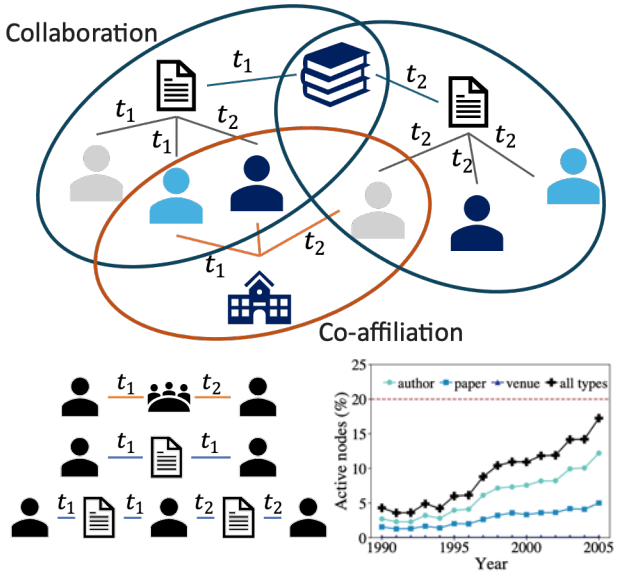
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Entity



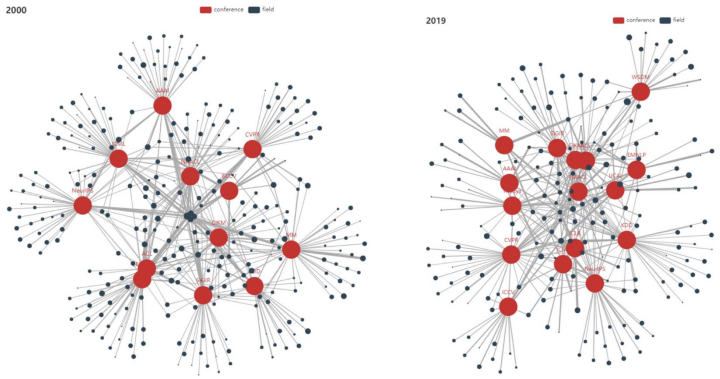
Temporal Heterogeneity

Motif



Semantic Evolution

Network



Distribution Shift

[1] Gao, Shen, et al. A Trend of AI Conference Convergence in Similarity: An Empirical Study Through Trans-Temporal Heterogeneous Graph. IEEE TKDE. 2023

Limitations

Several surveys focus on separate aspect:

- Heterogeneous Graph Learning
  - Task-oriented: [Shi et al., 2017; Xie et al., 2021]
  - Framework-oriented: [Yang et al., 2020; Wang et al., 2022]

- Dynamic Graph Learning
  - Architectures: [Kazemi et al., 2020; Zheng et al., 2025; Feng et al., 2026]
  - Techniques: [Barros et al., 2021; Jiao et al., 2025]

[1] Shi, Chuan, et al. A Survey of Heterogeneous Information Network Analysis. IEEE TKDE 2017.
[2] Xie, Yu, et al. A Survey on Heterogeneous Network Representation Learning. Pattern Recognition 2021.
[3] Yang, Carl, et al. Heterogeneous Network Representation Learning: A Unified Framework with Survey and Benchmark. IEEE TKDE 2020.
[4] Wang, Xiao, et al. A Survey on Heterogeneous Graph Embedding: Methods, Techniques, Applications and Sources. IEEE TBD 2022.
[5] Kazemi, Seyed Mehran, et al. Representation Learning for Dynamic Graphs: A Survey. JMLR 2020.
[6] Zheng, Yanping, et al. A Survey of Dynamic Graph Neural Networks. Frontiers of Computer Science 2025.
[7] Feng, ZhengZhao, et al. A Comprehensive Survey of Dynamic Graph Neural Networks: Models, Frameworks, Benchmarks, Experiments and Challenges. IEEE TKDE 2026.
[8] Barros, Claudio D. T., et al. A Survey on Embedding Dynamic Graphs. ACM CSUR 2021.
[9] Jiao, Pengfei, et al. A Survey on Temporal Interaction Graph Representation Learning: Progress, Challenges, and Opportunities. IJCAI 2025.

Limitations

Existing surveys have limitations:

- Separate treatment of heterogeneity and dynamics.
- Limited coverage of DHG methodologies.
- Lack of a unified taxonomy across temporal granularities.
- Insufficient discussion of DHG benchmarks and evaluation protocols.

Research gaps call for a unified and systematic DHG survey.

Definitions

A unified definition covers both **snapshot sequences** and **event streams**.

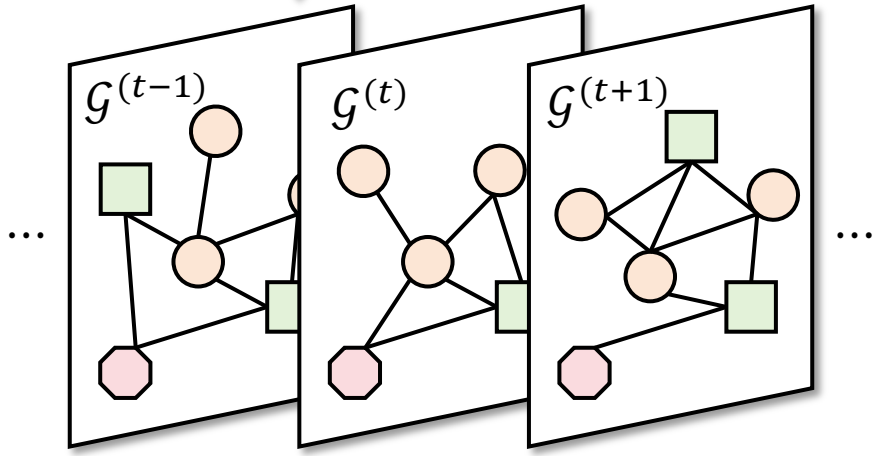
Definition 1 (Dynamic Heterogeneous Graph)**.** *A* Dynamic Heterogeneous Graph (DHG) *is defined as a graph jointly modeling structural heterogeneity and temporal evolution:*

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A}, \mathcal{R}, \mathcal{T}, \phi, \varphi),$$

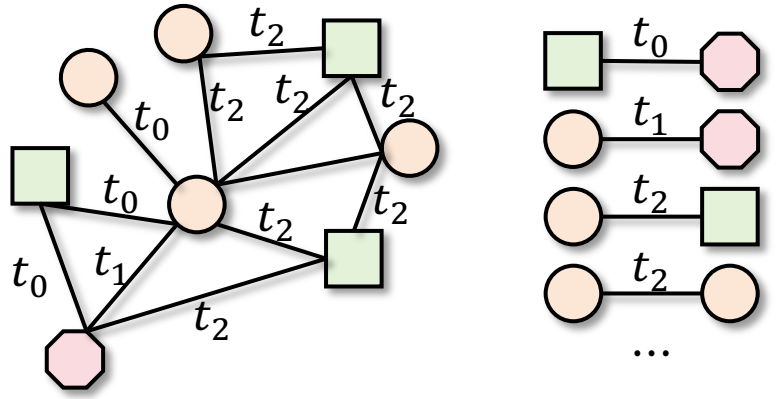
where $\mathcal{V}$ and $\mathcal{E}$ denote the (possibly time-varying) sets of nodes and edges over a temporal domain $\mathcal{T}$. The type-mapping functions $\phi : \mathcal{V} \to \mathcal{A}$ and $\varphi : \mathcal{E} \to \mathcal{R}$ assign each node and edge to a node-type set $\mathcal{A}$ and an edge-type set $\mathcal{R}$, respectively.

Regularly Grouping

Timestamp Preserving

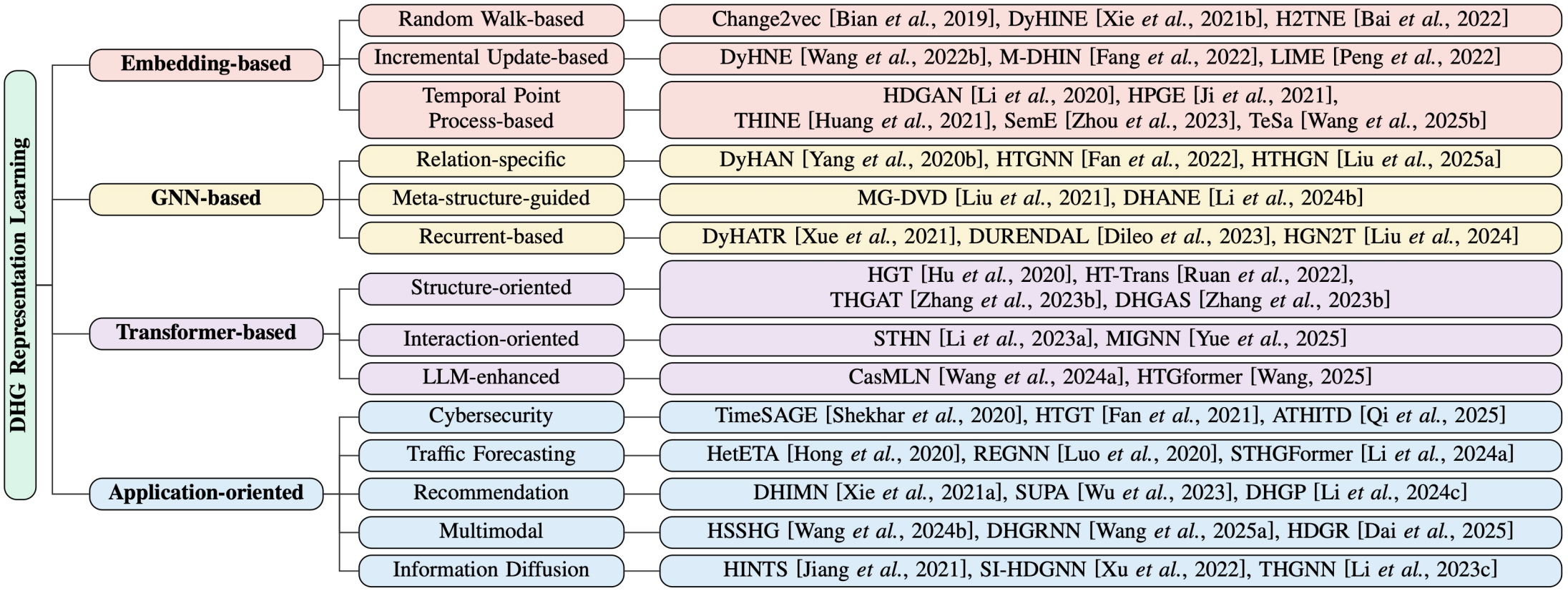


Discrete-time DHG $\mathcal{G} = \{\mathcal{G}^{(t)}\}_{t \in \mathcal{T}}$



Continuous-time DHG $\mathcal{E} = \{(u, v, r, t)\}_{t \in \mathcal{T}}$

Taxonomy



A Taxonomy of Dynamic Heterogeneous Graph Representation Learning

Summary

Category	Method	Granularity	Core Technique	Task	Code
Embedding-based	Change2vec [Bian *et al.*, 2019]	Dis.	Meta-path RW with Incremental Update	Clust.	Link
	THINE [Huang *et al.*, 2021]	Con.	Chronological Meta-path Walk with Hawkes Process	NC/LP/TLP	Link
	DyHINE [Xie *et al.*, 2021b]	Dis.	Relation-specific RW with Att.	TLP	-
	HPGE [Ji *et al.*, 2021]	Con.	Relation-specific Att. with TPP	NC/TLP	Link
	M-DHIN [Fang *et al.*, 2022]	Dis.	Meta-graph Sampling with LSTM	NC/LP/TLP	-
	DyHNE [Wang *et al.*, 2022b]	Dis.	Meta-path Proximity with Eigen Perturbation	NC/RP/TLP	Link
	LIME [Peng *et al.*, 2022]	Dis.	Meta-path RW with Recursive NN	NC/Clust.	Link
	SemE [Zhou *et al.*, 2023]	Con.	Semantic Meta-path Walk with Hawkes Att.	NC/RP/TLP	Link
	H2TNE [Bai *et al.*, 2022]	Con.	Relation-specific Walk with Hyperbolic Embedding	NC/LP/TLP	Link
	DHANE [Li *et al.*, 2024b]	Dis.	Meta-graph Att. with Online Update	NC	Link
GNN-based	HDGAN [Li *et al.*, 2020]	Con.	Meta-path GNN with TPP	NC/Clust.	-
	DyHAN [Yang *et al.*, 2020b]	Dis.	Relation-specific GNN with Temporal Att.	TLP	-
	DyHATR [Xue *et al.*, 2021]	Dis.	Relation-specific GNN with Temporal Att.	TLP	Link
	HTGNN [Fan *et al.*, 2022]	Dis.	Relation-specific GNN with Temporal Modeling	NR/TLP	Link
	DURENDAL [Dileo *et al.*, 2023]	Dis.	Relation-specific GNN with RNN	LP/TLP	Link
	HGN2T [Liu *et al.*, 2024]	Dis.	Relation-specific GNN with GRNN	LP/TLP	Link
	HTHGN [Liu *et al.*, 2025a]	Dis.	Relation-specific GNN with Temporal Att.	LP/TLP	Link
	TeSa [Wang *et al.*, 2025b]	Con.	Relation-specific GNN with TPP	LP/TLP	-
Transformer-based	HGT [Hu *et al.*, 2020]	Con.	Relation-specific Transformer with RTE	TLP	Link
	HT-Trans [Ruan *et al.*, 2022]	Con.	Relation-specific Transformer Encoder	LP/TLP	-
	DHGAS [Zhang *et al.*, 2023b]	Dis.	Relation- and Time-aware Self-Att.	NC/NR/RP/LP/TLP	Link
	THGAT [Zhang *et al.*, 2023a]	Con.	Relation-specific Att. with RTE	NC/Clust./TLP	Link
	STHN [Li *et al.*, 2023a]	Con.	Structural-Temporal Transformer	LP/TLP	Link
	CasMLN [Wang *et al.*, 2024a]	Dis.	LLM-enhanced Relation-specific Transformer	NC/NR/RP/LP/TLP	Link
	HTGformer [Wang, 2025]	Dis.	LLM-enhanced Transformer for DHG	NC/NR/RP/LP/TLP	-
	MIGNN [Yue *et al.*, 2025]	Con.	Att.-based GNN with GRU	LP/TLP	-
Application-oriented	HTGT [Fan *et al.*, 2021]	Dis.	Relation-specific Att. with RTE	Cybersecurity	Link
	ATHITD [Qi *et al.*, 2025]	Dis.	Relation-specific Att. with RTE	Cybersecurity	-
	HetETA [Hong *et al.*, 2020]	ST.	Heterogeneous GNN with Causal Conv.	Traffic	Link
	STHGFormer [Li *et al.*, 2024a]	ST.	Relation-specific Att. with Temporal Att.	Traffic	-
	SUPA [Wu *et al.*, 2023]	Con.	Meta-path RW with Memory Update	Recomm.	Link
	DHGP [Li *et al.*, 2024c]	Dis.	Relation- and Time-aware Att.	Recomm.	Link
	DHGRNN [Wang *et al.*, 2025a]	ST.	Relation-specific Att. with GRU	Multimodal	-
	HDGR [Dai *et al.*, 2025]	Dis.	Relation-specific GConv. with Reconstruction	Multimodal	-
	SI-HDGNN [Xu *et al.*, 2022]	Con.	Relation-specific RW with GRU	Diffusion	Link
	THGNN [Li *et al.*, 2023c]	Con.	Relation-specific Att. with LSTM	Diffusion	Link

Table 1: Comparison of DHG representation learning methods. Task abbreviations: NC (node classification), LP/TLP (link / temporal link prediction), RP (relation prediction), NR (node regression), Clust. (node clustering). Temporal granularity: Dis. (discrete-time), Con. (continuous-time), ST. (spatio-temporal). Technique abbreviations: RW (random walk), Att. (attention), GConv. (graph convolution), GRNN (graph recurrent neural network), TPP (temporal point process), RTE (relative time encoding).

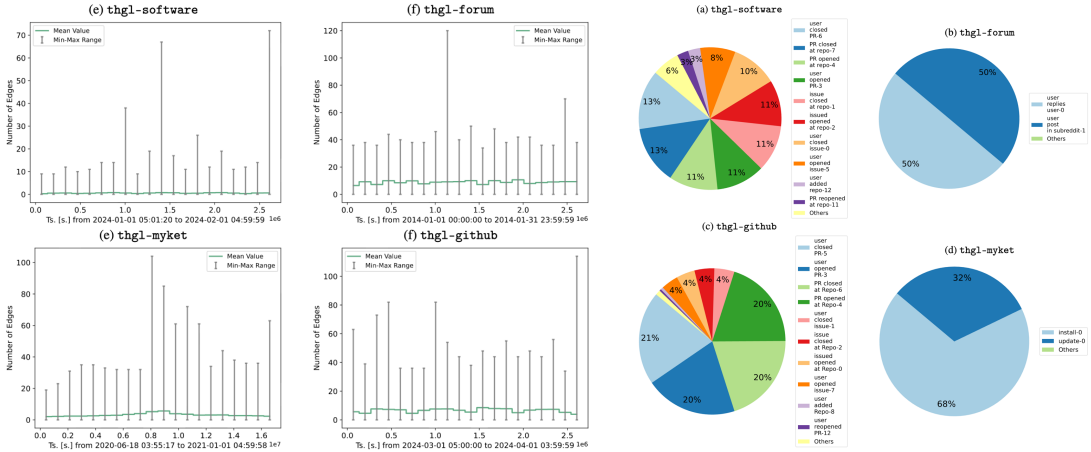
- Methods evolved from embeddings to GNNs and Transformers.
- Relation-aware GNNs and attention dominate applications.
- Most methods focus on temporal link prediction.

Datasets & Benchmarks

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	Domain	Dataset	$\lvert\mathcal{A}\rvert$	$\lvert\mathcal{R}\rvert$	$\lvert\mathcal{T}\rvert$	Label	URL
Datasets	Academic	DBLP	4	3	10	✓	Link
	Academic	AMiner	4	3	10	✓	Link
	Review	Yelp	4	3	10	✓	Link
	Social	Twitter	1	3	7	×	Link
	Epidemic	COVID	2	3	304	✓	Link
	Recomm.	Movielens	3	3	25,864	×	Link
	Recomm.	Ecomm	2	4	11	×	Link
	Academic	MAG	4	4	10	✓	Link
	Recomm.	Alibaba	2	4	11	×	Link
	Academic	Math-Overflow	1	3	11	×	Link
TGB 2.0	Software	software	4	14	689,549	×	Link
	Social	forum	2	2	2,558,457	×	Link
	Software	github	4	14	2,510,415	×	Link
	Recomm.	myket	2	2	14,828,090	×	Link

Dataset statistics



Dataset distribution

	Temporal Knowledge Graphs (tkgl-)				Temporal Heterogeneous Graphs (thgl-)			
Dataset	smallpedia	polecat	icews	wikidata	software	forum	github	myket
Domain	knowledge	political	political	knowledge	software	social.	software	interac.
# Quadruples	550,376	1,779,610	15,513,446	9,856,203	1,489,806	23,757,707	17,499,577	53,632,788
# Nodes	47,433	150,931	87,856	1,226,440	681,927	152,816	5,856,765	1,530,835
# Edge Types	283	16	391	596	14	2	14	2
# Node Types	-	-	-	-	4	2	4	2
# Timesteps	125	1,826	10,224	2,025	689,549	2,558,457	2,510,415	14,828,090
Granularity	year	day	day	year	second	second	second	second
Induct. Test Nodes	0.26	0.12	0.05	0.34	0.13	0.02	0.14	0.01
DRec	0.71	0.07	0.11	0.61	0.00	0.00	0.00	0.00
Rec	0.72	0.43	0.63	0.61	0.10	0.63	0.01	0.37
Con	5.82	1.07	1.14	5.05	1.00	1.00	1.00	1.00
Mean Edges/Ts.	4,403.01	974.59	1,516.91	4,867.26	0.56	8.87	6.54	3.15
Mean Nodes/Ts.	5,289.16	550.60	1,008.65	5,772.16	0.86	12.96	9.77	6.24

Benchmark

- Inconsistent construction and temporal discretization
- Non-standardized evaluation protocols
- Narrow task coverage and limited baselines

More comprehensive DHG benchmarks remain needed.

[1] Gastinger, Julia, et al. TGB 2.0: A Benchmark for Learning on Temporal Knowledge Graphs and Heterogeneous Graphs. NeurIPS 2024.

Future Direction

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- ✓ **Efficiency:** Essential for scaling DHG learning to **web-scale pre-train** while preserving heterogeneous semantics and temporal dynamics.

Table 3: **MRR** results for *Temporal Heterogeneous Graph Link Prediction* task. We report the average and standard deviation across 5 different runs. First place is **bolded** and second place is underlined.

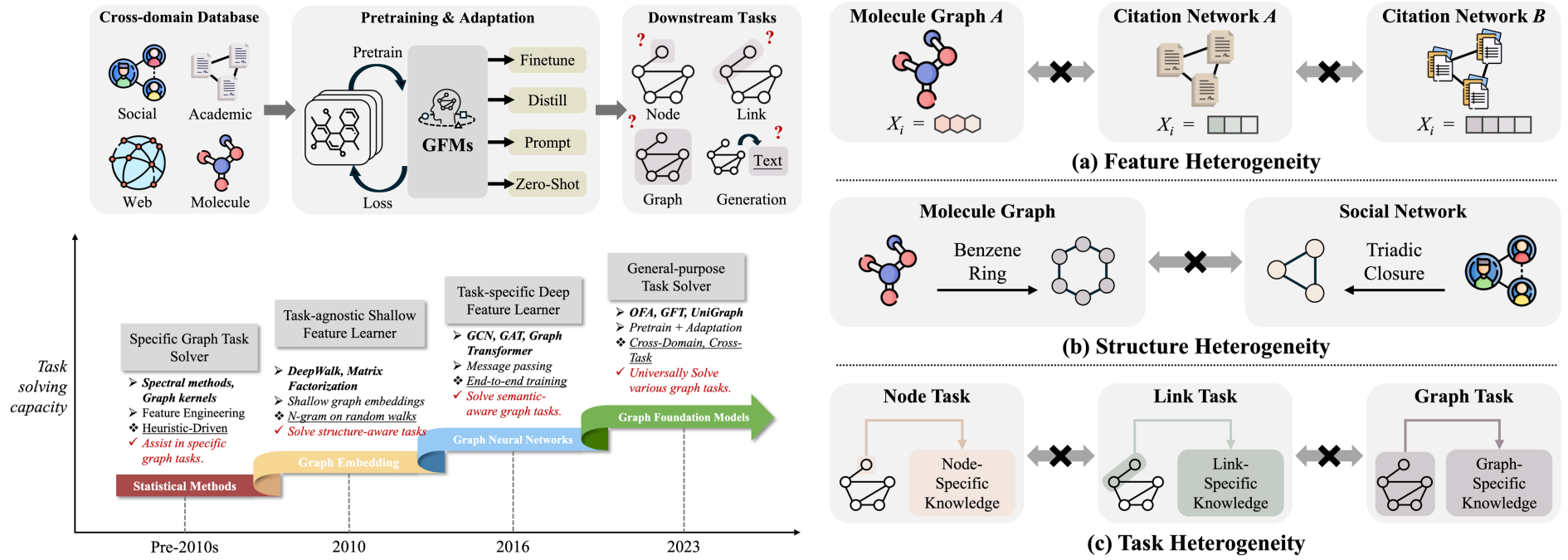
Method	thgl-software		thgl-forum		thgl-github		thgl-myket	
	Validation	Test	Validation	Test	Validation	Test	Validation	Test
EdgeBank$_{tw}$ [54]	0.279	0.288	0.534	0.534	0.355	0.374	0.248	0.245
EdgeBank$_{\infty}$ [54]	0.399	0.449	0.612	0.617	0.403	0.413	0.430	0.456
RecB$_{default}$ [13]	0.106	0.099	0.552	0.561	OOT	OOT	OOT	OOT
TGN [58]	0.299±0.012	0.324±0.017	0.598±0.086	0.649±0.097	OOM	OOM	OOM	OOM
TGN$_{edge-type}$	0.376±0.010	0.424±0.013	**0.767**±0.005	**0.729**±0.009	OOM	OOM	OOM	OOM
STHN [36]	**0.764**±0.025	**0.731**±0.005	OOM	OOM	OOM	OOM	OOM	OOM

Revealing scalability bottleneck

[1] Gastinger, Julia, et al. TGB 2.0: A Benchmark for Learning on Temporal Knowledge Graphs and Heterogeneous Graphs. NeurIPS 2024.

Future Direction

- ✓ **Generalizability:** Critical for enabling **adaption** DHG representations across **multi-tasks/domains**, and **data distributions**.



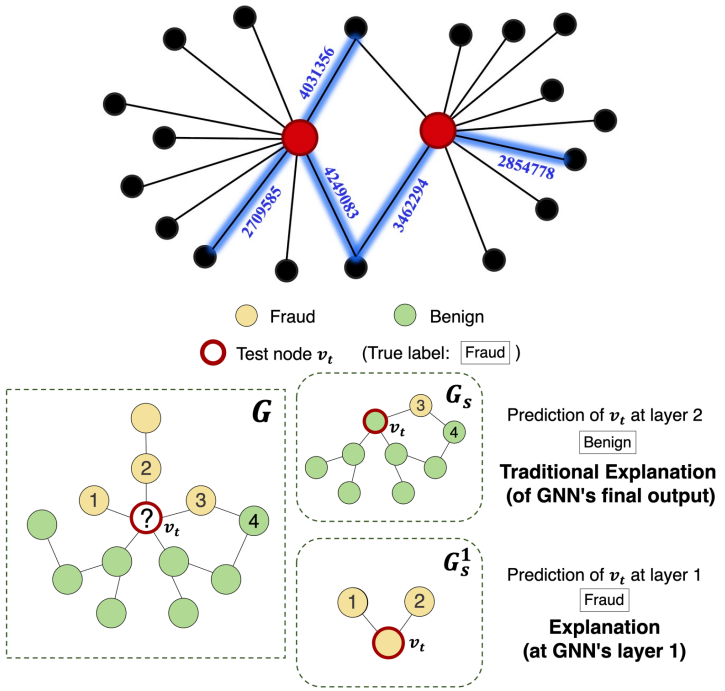
Lacking effective pre-training and adaptation

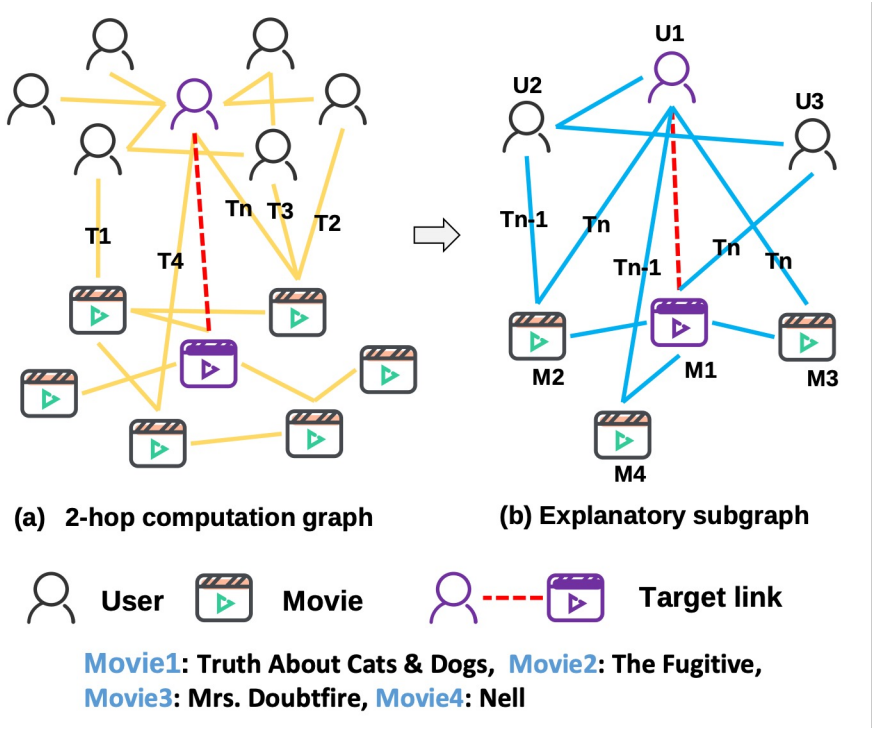
[1] Wang, Zehong, et al. Graph Foundation Models: Challenges, Methods, and Open Questions. KDD 2025.

Future Direction

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✓ **Trustworthiness:** Critical for the **explainability** and **interpretability** of DHG reasoning, particularly in high-stakes, time-sensitive domains.





Lacking groundtruth and reliable reasoning

[1] Chen, Hongjiang, et al. ST-TGExplainer: Disentangling Stability and Transition Patterns for Temporal GNN Interpretability. ICML 2026.
[2] Yu, Cibo, et al. SliceGX: Layer-wise GNN Explanation with Model-slicing. WWW 2026.
[3] Li, Jiazheng, et al. Heterogeneous Temporal Graph Neural Network Explainer. CIKM 2023.

Conclusion

- **Unified Perspective:**
  - Bridging typed semantics and temporal evolution
  - Unifying snapshot-based and event-driven DHGs
- **Novel Taxonomy:**
  - Reframing DHG methods by learning mechanisms
  - Revealing how models integrate type and time
- **Comprehensive Review:**
  - Connecting methods, applications, and benchmarks
  - Exposing shared bottlenecks and evaluation gaps
- **Future Directions:**
  - Scaling to Web-scale evolving graphs
  - Building transferable DHG foundation models
  - Enabling faithful and causal explanations

Thank You!

Questions & Discussion

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Lab Homepage